

SRS Executive Summary & Technical Brief

Zomato Clone - Food Ordering & Delivery Application

Project Name:	Zomato Clone (MERN Web App)	Developer / Intern:	Mohit Sharma
Senior Mentor:	Mentor Aryaman	Program:	Web Development Internship
SRS Standard:	IEEE Std 830-1998 Format	Date:	July 24, 2026

1. Purpose & High-Level Architecture

This document presents a concise technical summary of the complete 7-page Software Requirements Specification (SRS) for the **Zomato Clone** application. Developed by **Mohit Sharma** under the mentorship of **Mentor Aryaman**, the application reproduces commercial food ordering and delivery workflows using a modern, decoupled MERN stack architecture.

Architectural Tier	Technology Stack	Core Functionality & Responsibilities
Frontend (Port 5173)	React 18, Vite, Redux Toolkit, Tailwind CSS	Single Page Application (SPA) rendering home feed, search, location selector, restaurant menus, cart, checkout, and order history.
Backend (Port 5000)	Node.js, Express.js (MVC Pattern)	RESTful JSON API under `/api/v1`, JWT token authentication middleware, password hashing, business logic, and error handlers.
Database (Port 27017)	MongoDB & Mongoose ORM	BSON document store keeping data for `User`, `Restaurant`, `FoodItem`, and `Order` collections with schema validation.

2. Summary of Functional Requirements

Module	Core Capabilities & Processing Logic	Key Features / Rules
Authentication (FR-1)	User registration & login authentication. Bcrypt password encryption (10 salt rounds), JWT token issuance (7-day validity), and protected route guards.	- Demo user access (`demo@example.com`) enabled. - Protected checkout route guard.
Restaurant Browsing (FR-2)	Browse 50+ Indian restaurants. Client-side search, cuisine pills, 4.0+ rating filter, and delivery duration filters.	- Unsplash image fallbacks. - Loading skeleton state.
Location Engine (FR-2.2)	GPS Geolocation + OpenStreetMap Nominatim reverse & forward geocoding. Computes physical distance in km using Haversine formula.	- Closest-first sorting. - Distance tag pills (`2.4 km away`).
Menu & Badges (FR-2.3)	Categorized dish catalogs (Starters, Mains, Desserts, Beverages). Renders Veg/Non-Veg indicators & promotional badges.	🔥 Bestseller ★ Top Rated 👨🍳 Chef's Special
Cart & Pricing (FR-3)	Redux Toolkit cart management. Dynamic computation of Subtotal, flat ₹45 delivery fee (waived over ₹500), and 5% GST tax.	- Single-kitchen cart guard rule. - Dynamic total calculations.
Checkout & Orders (FR-4)	Order placement (POST `/api/v1/orders`), address assignment, payment options (COD/Card/UPI), confirmation screen, and status tracking.	Status tracking lifecycle (`Placed`, `Preparing`, `Delivered`).

3. Database Schema Overview

The database contains four core collections managed via Mongoose schemas:

Collection Model	Key Schema Fields & Relationships	Operational Purpose
`User`	`name`, `email` (unique), `password` (hashed, select: false), `role` ('customer' 'partner' 'admin'), `addresses` array.	Stores user profiles, roles, and saved delivery addresses.
`Restaurant`	`name`, `image`, `cuisine` array, `rating`, `deliveryTime`, `costForTwo`, `address`, `coordinates` (`lat`, `lon`), `isFeatured`, `partner` (ref User).	Stores restaurant listings, city addresses, and GPS coordinate nodes.
`FoodItem`	`name`, `description`, `price`, `category`, `isVeg`, `isBestselling`, `isTopRated`, `isTodaysSpecial`, `image`, `restaurant` (ref Restaurant).	Stores dish catalog (~2,500 items) with promotional tag flags.
`Order`	`user` (ref User), `restaurant` (ref Restaurant), `items` array (`foodItem`, `name`, `price`, `quantity`), `totalAmount`, `deliveryAddress`, `paymentMethod`, `status`.	Records customer purchase transactions and delivery status.

4. Non-Functional Requirements & Security Summary

- **Performance Targets:** Vite bundle compilation under 325 kB; initial DOM load under 1.5 seconds; database response under 50ms.
- **Security Architecture:** Bcrypt password encryption (10 salt rounds); JWT token validation middleware; MongoDB ObjectID sanitization.
- **Reliability & Error Safety:** Server health check endpoint (`GET /api/v1/health`); centralized Express error middleware returning structured JSON.
- **Maintainability:** Strict MVC backend architecture; sliced Redux Toolkit state stores (`authSlice`, `cartSlice`, `restaurantSlice`).

5. Document Sign-Off & Verification

Role / Title	Name	Status / Signature	Date
Developer / Intern	Mohit Sharma	Submitted for Evaluation	July 24, 2026
Senior Mentor	Mentor Aryaman		July 24, 2026